Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-D Valvular Heart Diseases

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-04-C01				
NPLEX-P1-CARD-18-04-C02				
NPLEX-P1-CARD-18-04-C03				
NPLEX-P1-CARD-18-04-C04				
NPLEX-P1-CARD-18-04-C05				
NPLEX-P1-CARD-18-04-C06				
NPLEX-P1-CARD-18-04-C07				
NPLEX-P1-CARD-18-04-C08				
NPLEX-P1-CARD-18-04-C09				
NPLEX-P1-CARD-18-04-C10				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C01-BD1

Rebuild calcific aortic stenosis and the wall it builds

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C01 Activity: NPLEX-P1-CARD-18-04-C01-BD1 Date: _____

Without looking at your notes, write the normal aortic valve area and the area at which symptoms usually begin. Then write out the steps of age-related calcific aortic stenosis from the first endothelial injury to the heaped calcified masses, naming what enters the cusp, which cell changes its behaviour, and what that cell starts to deposit. State what the serum calcium is in this process and what the commissures look like. Beside that, write the two other causes of aortic stenosis, the typical age of each, and the one feature that identifies the rheumatic valve. Next, write the law of Laplace as a relationship between wall stress, pressure, radius, and wall thickness, and use it to explain why the ventricle thickens around a normal cavity rather than dilating. Name that pattern of hypertrophy, write what happens to ejection fraction for the first years, and write which heart sound appears once the wall is stiff and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C01-BD2

Work a Doppler study back to the symptoms

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C01 Activity: NPLEX-P1-CARD-18-04-C01-BD2 Date: _____

A woman of 76 years has an aortic jet velocity of 4.6 m/s, a valve area of 0.8 cm2, and an arterial pressure of 128/76 mm Hg. Convert the jet velocity to a peak gradient using the simplified Bernoulli relationship, show your working, and state the pressure the left ventricle must generate. Using Laplace with an unchanged cavity radius, calculate how thick a wall that began at 1.0 cm must become to hold wall stress steady, and show the ratio you used. State whether this study meets the thresholds for severe disease and give each threshold. Then explain, in two or three sentences each, why she has angina even though her coronary arteries are clean, why her syncope comes only on exertion, and why she is breathless. Write what her carotid upstroke and apical impulse feel like and why. Finish by writing what happens to the loudness of her murmur if the ventricle begins to fail, and why loudness is a poor guide to severity.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-04-C01-Q01

Chest pain with clean coronary arteries

A 74-year-old man comes to the physician because of 6 months of chest tightness on climbing stairs and one episode of collapse while running. Temperature is 36.8 °C (98.2 °F), pulse is 72/min, and blood pressure is 126/80 mm Hg. The carotid upstroke is weak and late and the apical impulse is sustained. A crescendo-decrescendo systolic murmur is loudest in the right second intercostal space and radiates to both carotids, A2 is soft, and an S4 is present. Coronary angiography shows no epicardial stenosis. Echocardiography shows a valve area of 0.8 cm2, a mean gradient of 48 mm Hg, and a left ventricular wall thickness of 1.6 cm with an ejection fraction of 60%. Which of the following best explains this patient's chest pain?

A. Thrombotic occlusion of a coronary artery over a ruptured atherosclerotic plaque

B. Embolism of a calcified valve fragment into a coronary artery

C. High oxygen demand in a thick wall whose raised diastolic pressure compresses subendocardial vessels while prolonged systole shortens diastolic perfusion time

D. Reduced oxygen carrying capacity from loss of large von Willebrand factor multimers

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C02-BD1

Causes of aortic regurgitation and the wide pulse pressure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C02 Activity: NPLEX-P1-CARD-18-04-C02-BD1 Date: _____

Divide the causes of aortic regurgitation into root disease and leaflet disease and list every cause the notes gave under each heading. For Marfan syndrome, name the mutated protein, the growth factor whose activity rises, and the name given to the change in the aortic media. For tertiary syphilis, write which vessels are inflamed, what that does to the media, and what the intima looks like. Then write what the left ventricle receives in diastole in chronic aortic regurgitation, which mechanism raises total stroke volume, and name and describe the pattern of hypertrophy that follows. Write why diastolic pressure in the ventricle stays near normal for years. Next, give a representative blood pressure in chronic aortic regurgitation and write the two reasons diastolic arterial pressure falls. List the named peripheral signs from the notes and write the one shared finding that produces all of them. Finish by writing the murmur: its pitch, its timing, where it starts in the cardiac cycle and why, where and in which position it is heard best, and what handgrip does to it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C02-BD2

Two patients with a leaking aortic valve

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C02 Activity: NPLEX-P1-CARD-18-04-C02-BD2 Date: _____

Patient 1 has had a dilated aortic root for 15 years, feels well, and has a blood pressure of 172/44 mm Hg with bounding pulses and a long early diastolic murmur. Patient 2 presents within an hour of an ascending aortic dissection with a blood pressure of 92/68 mm Hg, pulmonary edema, and a short soft diastolic murmur. For each patient write the size and stiffness of the left ventricle, what happens to left ventricular diastolic pressure, the length of the murmur and why, and whether the pulse pressure is wide, explaining each answer from the compliance of the chamber receiving the regurgitant volume. Write why patient 2 can have severe regurgitation and almost no murmur, and what the rising ventricular pressure does to the mitral valve. Then write what a fall in heart rate does to the regurgitant volume per beat in each patient and why. Finish by writing how you would separate the apical rumble of an Austin Flint murmur from mitral stenosis at the bedside, naming the two findings mitral stenosis adds.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C03-BD1

Mitral stenosis: the chambers that pay and the sounds you hear

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C03 Activity: NPLEX-P1-CARD-18-04-C03-BD1 Date: _____

Write the normal mitral valve area, the area at which a gradient first appears, and the area that counts as severe. Name the usual cause and the two uncommon ones. Then write what happens to the size and pressure of the left atrium and to the size and diastolic pressure of the left ventricle, and write the one haemodynamic measurement that identifies mitral stenosis when compared with left ventricular diastolic pressure. Explain why hemoptysis occurs. Next, list from memory the findings of mitral stenosis with the mechanism of each: the character of S1, the extra diastolic sound and what makes it, what happens to the interval between S2 and that sound as the disease worsens and why, the murmur with its pitch, timing, where and in which position it is heard, which feature of the murmur tracks severity, the accentuation before systole and what abolishes it, and the two electrocardiographic patterns. Finish by writing the three structures an enlarged left atrium can press on and the symptom each produces.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C03-BD2

Why a faster heart rate floods the lungs

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C03 Activity: NPLEX-P1-CARD-18-04-C03-BD2 Date: _____

A woman of 32 years with known mitral stenosis has been well for years. At 28 weeks of pregnancy she develops a fever of 38.6 °C (101.5 °F) and a heart rate that rises from 72/min to 124/min, and within hours she has pulmonary edema. Write the relationship between flow across a narrowed orifice, the orifice area, and the pressure difference across it, then explain why a rise in heart rate raises left atrial pressure so sharply, naming what tachycardia does to diastole. Estimate what happens to the gradient at the new heart rate and say what the notes give for a rise of this size. List every factor in this patient that pushed her over, and write what pregnancy itself contributed. Then write her left ventricular end-diastolic pressure and ventricular size and justify each from where the obstruction sits. Next, she develops atrial fibrillation. Write the two ways that worsens filling, where thrombus forms and why, and name four organs the emboli reach. Finish by writing how her picture differs from pulmonary edema caused by left ventricular failure, using the wedge pressure and the ventricular diastolic pressure to make the distinction.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-04-C03-RT01

Rebuild the backward transmission of pressure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C03 Activity: NPLEX-P1-CARD-18-04-C03-RT01 Date: _____

Rebuild the seven-step sequence that runs from the narrowed mitral orifice to right heart failure, one step per line and in order. Beside the left atrial step, write what happens to the wall of the atrium and what happens to the left ventricle at the same time. Beside the pulmonary venous step, write the three symptoms that follow. Beside the pulmonary arterial step, write what the arterioles do first and what then changes in their walls. Beside the right ventricular step, name the valve lesion that appears and why. Beside the final step, write the three physical findings you expect. Then draw a vertical bar for every chamber and vessel from the pulmonary veins to the left ventricle and mark with an arrow which pressures are raised and which are not, and write one sentence on why the ventricle is the one chamber spared.

Rebuild from memory: Backward transmission of pressure in mitral stenosis

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C04-BD1

The mitral apparatus and how each part fails

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C04 Activity: NPLEX-P1-CARD-18-04-C04-BD1 Date: _____

Name the five parts of the apparatus that keep the mitral valve competent. Then separate the causes of mitral regurgitation into primary and secondary, list the causes the notes gave under each, and name the most common primary cause in developed countries. Write which papillary muscle ruptures most often after an infarction and the anatomical reason. For secondary regurgitation, write what a dilated ventricle does to the annulus and to the papillary muscles, and explain in two sentences why this form feeds itself. Next, write why backflow begins at the very onset of systole and what that does to isovolumetric contraction. Write which chambers carry the extra volume and when the regurgitant volume returns to the ventricle. Describe what happens to the size and compliance of the left atrium and the rhythm it often develops, then name the pattern of ventricular hypertrophy. Finish with the murmur and sounds: timing, pitch, where it is loudest, where it radiates, what handgrip and squatting do, what standing and Valsalva do, the character of S1, what happens to the split of S2, and which extra sound the loaded atrium produces and what it does and does not prove.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C04-BD2

Why a normal ejection fraction misleads

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C04 Activity: NPLEX-P1-CARD-18-04-C04-BD2 Date: _____

A man with chronic mitral regurgitation has an end-diastolic volume of 210 mL and an end-systolic volume of 75 mL, and 60% of his total stroke volume returns to the left atrium. Calculate his total stroke volume, his ejection fraction, his regurgitant volume, and his forward stroke volume, showing each step. Then repeat the calculation with contractility reduced so that end-systolic volume rises to 105 mL with the same filling and the same regurgitant fraction, and state the new ejection fraction and forward stroke volume. Write two or three sentences explaining why an ejection fraction a clinician might call low-normal already reports a failing ventricle in this lesion, naming the feature of the outlet that flatters it. Then compare this with aortic stenosis, where the ventricle ejects against a high pressure, and write what that difference does to the usefulness of ejection fraction. Finish by writing which single measurement in this patient would tell you how much blood is actually reaching his tissues.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-04-C04-CP01

Chronic against acute regurgitation

Fill in every cell from memory for chronic and for acute regurgitation, covering both the aortic and the mitral lesion where the row allows it. Then check your table against the notes and correct any cell you missed.

Feature	Chronic regurgitation	Acute regurgitation
Size and compliance of the chamber receiving the backflow		
What happens to filling pressure, and which pressure you would name in aortic and in mitral disease		
Length and loudness of the murmur, named for the aortic and the mitral lesion		
Pulse pressure in aortic regurgitation		
How the patient presents and over what time		
Usual causes in aortic and in mitral disease		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-04-C04-CA01

Sudden breathlessness four days after an infarction

A 64-year-old man is on the cardiology ward on the fourth day after an inferior myocardial infarction that was treated late. He becomes acutely breathless over 20 minutes. Temperature is 36.9 °C (98.4 °F), pulse is 118/min, respirations are 32/min, and blood pressure is 84/60 mm Hg. Oxygen saturation is 86% on room air. There are coarse crackles throughout both lung fields. A short, soft systolic murmur is just audible at the apex. Chest radiography shows pulmonary edema with a normal cardiac silhouette. A pulmonary artery catheter shows a wedge pressure of 34 mm Hg with tall v waves. Echocardiography shows a left atrium of normal size, a left ventricle of normal size with inferior wall hypokinesis, and a flail segment of the posterior mitral leaflet.

1. Recognize. Name the valve lesion, state whether it is acute or chronic, and name the structure that has failed and the part of the apparatus it belongs to.

2. Explain. Explain why the pulmonary edema and the tall v waves appeared within minutes rather than over years, using the size and compliance of the receiving chamber.

3. Differentiate. Write how this presentation differs from chronic mitral regurgitation and from acute aortic regurgitation, naming the findings on examination and on the catheter tracing that separate them.

4. Investigate. Write which findings you would look for on the echocardiogram and the catheter tracing to confirm the lesion, and what each one would show.

5. Decide. State why the murmur is soft and short here even though the regurgitant volume is large, and what you would conclude about severity from its loudness.

6. Justify. Justify your reasoning from the vascular supply of the structure that ruptured, the territory of the infarct, and the pressure relationship between the left atrium and the left ventricle late in systole.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C05-BD1

How a vegetation forms and what the organism decides

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C05 Activity: NPLEX-P1-CARD-18-04-C05-BD1 Date: _____

Write what a vegetation is made of and explain in two sentences why organisms growing inside it survive an intact immune system. Then rebuild the six-step pathogenesis of infective endocarditis in order, one step per line, and write beside the first two steps which organism can skip them and what that allows it to do. List the conditions that expose collagen and tissue factor on a valve, and list the everyday events that deliver a transient bacteremia. Name the surface molecules by which viridans streptococci and Staphylococcus aureus attach. Next, build a two-column comparison of acute and subacute disease with a row each for the usual organism, the state of the valve before infection, the time course, the fever and systemic features, the size and destructiveness of the vegetation, and whether embolic or immune signs dominate, and write one sentence on why virulence sets the tempo. Finish by writing which valve is most often involved in native valve disease, which valve is most often involved in people who inject drugs and why, where right-sided emboli travel and what they look like, and four places left-sided emboli lodge.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C05-BD2

Read the peripheral signs and the laboratory panel

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C05 Activity: NPLEX-P1-CARD-18-04-C05-BD2 Date: _____

A woman of 58 years has had 7 weeks of low-grade fever, fatigue, and weight loss. She has a repaired bicuspid aortic valve, painful red-purple nodules on three fingertips, painless flat red macules on both palms, retinal hemorrhages with pale centres, hematuria with red cell casts, an erythrocyte sedimentation rate of 86 mm/h, a hemoglobin of 9.4 g/dL, and a positive rheumatoid factor. Two sets of blood cultures drawn 8 hours apart grow the same organism. For each physical finding, name it, state whether it is painful, and write its mechanism, then write the single rule that lets you sort the painful from the painless signs. Explain why her urine has red cell casts and her complement is low, and why her prolonged illness favours this mechanism. Explain why her anemia developed and name the hormone that drives it. Write why both culture sets grew the same organism and what that tells you about the vegetation. Name the organism group you expect from this tempo and this valve. Finish by listing which of her findings count as major and which as minor by the Duke criteria, and give the temperature threshold the criteria use in Celsius with Fahrenheit in brackets.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-04-C05-Q02

A painful nodule on the finger pad

A 61-year-old woman comes to the physician because of 6 weeks of low-grade fever, fatigue, and a 4-kg weight loss. She has rheumatic mitral valve disease. Temperature is 37.9 °C (100.2 °F). Examination shows tender red-purple nodules on the pads of two fingers, splenomegaly, and a holosystolic apical murmur. Urinalysis shows hematuria with red cell casts, serum complement is low, and rheumatoid factor is positive. Blood cultures drawn at separate times grow viridans streptococci. Which of the following best explains the tender nodules on her finger pads?

A. Immune complex deposition with small vessel vasculitis in the digital pad

B. Septic microemboli lodging in dermal capillaries with small abscesses

C. Fibrin and platelet plugs occluding retinal arterioles

D. Dilation of a digital artery wall seeded by circulating organisms

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C06-BD1

The two sterile vegetations

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C06 Activity: NPLEX-P1-CARD-18-04-C06-BD1 Date: _____

Write the full name of nonbacterial thrombotic endocarditis and the other name it goes by, then describe its vegetations: size in millimetres, what they are made of, how firmly they are attached, which valves and which part of the leaflet they sit on, and how much inflammation surrounds them. Write the one condition it is most associated with, naming the tumour types, and write the two tumour products that drive the clotting and the vein condition they also cause. List the other settings that produce it. Then write whether a murmur is usual, where the fragments travel, and the presentation in a patient with cancer that should bring it to mind. Write one further thing an existing sterile lesion can become. Next, write the disease in which Libman-Sacks endocarditis occurs and the antibody often found with it, then describe its vegetations: size, appearance, which leaflet surfaces and which other structures they occupy, the histological change they show, and the valve lesion their healing most often produces. Finish by writing the single positional feature that points to Libman-Sacks rather than the other vegetations in this chapter.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C06-BD2

Strokes in three territories with negative cultures

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C06 Activity: NPLEX-P1-CARD-18-04-C06-BD2 Date: _____

A man of 67 years with pancreatic adenocarcinoma has three strokes in two weeks, in the territories of the right middle cerebral, left posterior cerebral, and left middle cerebral arteries. He has no fever and no murmur, and three sets of blood cultures are negative. Echocardiography shows small nodules along the closure lines of the mitral and aortic leaflets. Write which lesion this is and the two findings in his history and imaging that point to it. Explain the mechanism that put those nodules there, naming the tumour products involved and why this tumour type in particular. Explain why infarcts appear in unrelated arterial territories and why he has no murmur. Then write why negative cultures do not prove a vegetation is sterile, naming the two situations that produce a culture-negative infection, and list the findings that would push you back toward infection and the findings that keep you with a sterile lesion. Finish by writing what would change in the appearance and the behaviour of these nodules if organisms seeded them.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-04-C06-CP02

Four kinds of valve vegetation

Fill in every cell from memory, then check your table against the notes and correct any cell you missed.

Feature	Infective endocarditis	Nonbacterial thrombotic endocarditis	Libman-Sacks endocarditis	Acute rheumatic fever
Size and consistency of the vegetation				
Position on the valve and whether it spreads beyond the leaflet				
Organisms present or absent				
Degree of inflammation and the damage left behind				
Nature of the emboli				
Clinical setting in which it arises				

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C07-BD1

Myxomatous degeneration and the mid-systolic click

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C07 Activity: NPLEX-P1-CARD-18-04-C07-BD1 Date: _____

Define mitral valve prolapse in terms of what the leaflets do and where they end up, and give the proportion of adults affected. Name the underlying lesion, then write what accumulates in the spongiosa, what happens to the collagen of the fibrosa, and which cell drives the change and through which products. Describe what the leaflets, the chordae, and the annulus look like, and name the leaflet most often involved. Name the three inherited disorders the notes link to it and write the matrix defect in each. Next, explain in steps why a click appears in mid systole: write what the papillary muscles and annulus do as the ventricle empties, what the redundant leaflets then do, what the chordae do at the end of it, and where the sound is heard. Write what follows the click if the leaflets no longer meet, and the timing of that murmur. Finish by listing the complications the notes give and writing one line on how the weak tissue produces each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C07-BD2

Tell four systolic findings apart at the bedside

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C07 Activity: NPLEX-P1-CARD-18-04-C07-BD2 Date: _____

Write the rule that links left ventricular size to the timing of the click in mitral valve prolapse and the length of the murmur that follows it. List the manoeuvres that make the ventricle smaller and those that make it larger, and state what each group does to the click and to the murmur. Then four patients each have a systolic finding at the apex or the base: one has mitral valve prolapse, one has hypertrophic cardiomyopathy, one has aortic stenosis, and one has chronic mitral regurgitation. For each, write what standing and the strain phase of Valsalva do to the murmur, what squatting and sustained handgrip do, and the reason in one sentence drawn from preload, afterload, or chamber size. Write which two of the four behave the same way and the finding that still separates them. Then write the two differences between the mid-systolic click of prolapse and the opening snap of mitral stenosis, covering when in the cycle each occurs and which variable moves each one. Finish by writing which of these patients is at risk of sudden severe regurgitation and what has to happen for it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-04-C07-PR01

Predict what happens when a patient stands up

A patient with mitral valve prolapse and mild regurgitation moves from squatting to standing. Predict each item below, write the direction of the change, and give the reason in one line drawn from preload, ventricular size, or afterload.

What changes	Up, down, or no change	Why, in one or two lines
What happens to venous return and to left ventricular end-diastolic volume		
What happens to the size the ventricle reaches during systole, and how soon the leaflets lose support		
What happens to the timing of the mid-systolic click relative to S1		
What happens to the length of the late systolic murmur		
What would happen instead to the murmur of hypertrophic cardiomyopathy, and why		
What would happen instead to the murmur of aortic stenosis, and why		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C08-BD1

The chronic rheumatic valve and which valves it takes

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C08 Activity: NPLEX-P1-CARD-18-04-C08-BD1 Date: _____

Write the interval between group A streptococcal pharyngitis and acute rheumatic fever. Name the streptococcal antigen most involved, the two host tissues the immune response cross-reacts with, the name of that process, and the inflammatory lesion found in the heart during the acute illness. Then write why a single attack may leave little damage and what each new streptococcal throat infection does to an already scarred valve, naming the two tissue changes that let damage accumulate and explaining why the disease becomes evident only years to decades later. Next, write the four changes that define the chronic lesion. Describe what the stenotic mitral valve looks like seen from the atrium, and write why Aschoff bodies are rarely found at this stage. Write which of the four changes causes stenosis and which two prevent closure, and explain in two sentences why stenosis and regurgitation so often occur together here when degenerative disease usually causes one lesion alone. Finish with the valve distribution: the proportion with mitral disease alone, the proportion with mitral and aortic disease, what happens to the tricuspid and pulmonic valves, the usual explanation for the left-sided preference, and the one thing isolated aortic disease in an older adult should make you think of instead.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C08-BD2

Two narrowed valves with different histories

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C08 Activity: NPLEX-P1-CARD-18-04-C08-BD2 Date: _____

Patient 1 is 34 years old, grew up where streptococcal pharyngitis went untreated, and has a tight mitral valve with a loud S1, an opening snap, and a diastolic rumble, together with an aortic murmur. Patient 2 is 78 years old and has a tight aortic valve with a soft A2, pulsus parvus et tardus, and a normal mitral valve. For each patient write the likely cause of the valve disease and the two features of the valve itself, visible to a pathologist, that would confirm it. Write why the finding of an unaffected mitral valve in patient 2 effectively rules out rheumatic disease. Write what the commissures look like in each. Then write why patient 1 may have both narrowing and leakage of the same valve, naming the structural change behind each. Write which infection patient 1 is now at particular risk of, which organism group causes it, and why her valve invites it. Finish by writing why patient 1 is likely to decompensate during pregnancy, naming the three changes pregnancy brings and the one feature of her lesion that makes each of them dangerous.

My response

Confidence before checking

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After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-04-C08-RT02

Rebuild the path from a sore throat to a fused valve

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C08 Activity: NPLEX-P1-CARD-18-04-C08-RT02 Date: _____

Rebuild the seven-step sequence that runs from group A streptococcal pharyngitis to the chronic rheumatic valve, one step per line and in order. Beside the second step, write the antigen and the cross-reacting host tissues. Beside the third step, write where on the leaflet the vegetations sit and their size. Beside the fourth step, write the two healing changes and why one of them matters in a tissue that is normally avascular. Beside the fifth step, write what each recurrence adds. Beside the sixth step, write what happens to the commissures and to the chordae. Beside the final step, write what the orifice looks like and which other valve is commonly involved. Then write one sentence stating what the chronic disease actually is, so that it is clear the valve is not still infected.

Rebuild from memory: From streptococcal pharyngitis to chronic rheumatic valve disease

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C09-BD1

Serotonin, two filters, and the right-sided plaques

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C09 Activity: NPLEX-P1-CARD-18-04-C09-BD1 Date: _____

Write what kind of tumour a carcinoid is, the cell it arises from, and its most common site. Write the amino acid serotonin is made from and name the two enzymes in that pathway, then name the enzyme that breaks serotonin down, the metabolite produced, and where that metabolite is measured. List the other mediators these tumours release and name the peptide that one of them generates. Next, rebuild in seven steps the sequence from a serotonin-secreting bowel tumour to fibrous plaques on the right-sided valves. Using that sequence, explain why a tumour confined to the bowel usually causes no valve disease, why liver metastases change that, and how the lung protects the left heart, naming the two things pulmonary endothelium does to serotonin. Write the three situations in which left-sided lesions appear and the one tumour site that causes the syndrome without liver metastases, with the reason. Then describe the lesions: their colour, the cells and matrix they contain, which surfaces they lie on, and whether the valve beneath is destroyed. Name the receptor involved, the cell it acts on, the growth factor that helps, and two drug classes that produce similar plaques. Finish by writing what happens to the tricuspid valve and to the pulmonic valve, naming the commonest lesion of each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C09-BD2

A flushing patient with a murmur that changes with breathing

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C09 Activity: NPLEX-P1-CARD-18-04-C09-BD2 Date: _____

A man of 59 years has 8 months of episodic flushing, wheezing, and watery diarrhea with cramps, and a scaly rash with poor memory. He has a holosystolic murmur at the left lower sternal border that gets louder when he breathes in, large jugular v waves, a pulsatile liver, ankle edema, and ascites, plus a systolic ejection murmur at the left upper sternal border. For each symptom and sign write the mediator or the lesion that produces it, and name the valve lesion behind each of the two murmurs. Explain why the first murmur gets louder on inspiration. Explain the rash and the memory change, naming the vitamin that is short, the amino acid diverted away from making it, and the three features of that deficiency. Write the single urine test you expect to be abnormal and what is measured. Then write where this tumour must have spread for the heart to be involved and the reason, and write what you would predict about his left-sided valves and why. Finish with a second patient who has a bronchial carcinoid, no liver metastases, and a new valve lesion: write which side of the heart is affected and justify it from the route his tumour products take.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-04-C09-Q03

Valve disease in a patient with liver metastases

A 56-year-old woman comes to the physician because of 9 months of episodic flushing, wheezing, and watery diarrhea, and 3 months of ankle swelling. Examination shows a pulsatile liver, large jugular venous v waves, ascites, and a holosystolic murmur at the left lower sternal border that increases on inspiration. Urinary excretion of 5-hydroxyindoleacetic acid over 24 hours is markedly raised. Imaging shows an ileal mass with multiple liver metastases. Echocardiography shows pearly white plaques on the endocardial surfaces of the right-sided valves with normal left-sided valves. Which of the following best describes the valve lesions expected in this patient?

A. Mitral regurgitation from warty vegetations on both surfaces of the leaflets

B. Aortic stenosis from heaped calcified cusps with unfused commissures

C. Mitral stenosis from commissural fusion with a rigid fish-mouth orifice

D. Tricuspid regurgitation with pulmonic stenosis from fibrous plaques that fix the leaflets

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C10-BD1

Two kinds of prosthesis and the ways each one fails

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C10 Activity: NPLEX-P1-CARD-18-04-C10-BD1 Date: _____

Write what a mechanical valve is usually made of, how long it lasts, and the property of its surface that causes most of its trouble. Write what a bioprosthetic valve is made of, what glutaraldehyde does to the tissue and why that is done, and the consequence of the tissue being nonviable, giving the usual time to structural failure and the group in whom it comes sooner. Write what a transcatheter valve is. Then set out Virchow's triad and write what satisfies each of the three parts in a patient with a mechanical valve. Write which position carries the greater thrombotic risk and give two reasons. Write the two things a thrombus on the valve can do and the findings that follow when a leaflet is immobilised, then name the slower fibrous process that narrows the orifice and say where it grows from and over what time. Next, write what high-velocity turbulent flow does to red cells, name the defect that makes it worst and define it, and list every laboratory finding of that hemolysis with the reason for each. Write why chronic iron deficiency can follow. Finish by writing what happens to bioprosthetic tissue over time, the chemistry behind it, the group in whom it is faster, and the two ways the cusps then fail.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C10-BD2

Read the panel and name the complication

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C10 Activity: NPLEX-P1-CARD-18-04-C10-BD2 Date: _____

A woman with a mechanical mitral valve has 3 months of fatigue and dark urine. Hemoglobin is 8.8 g/dL, the blood film shows schistocytes including helmet cells, lactate dehydrogenase is high, unconjugated bilirubin is raised, haptoglobin is undetectable, reticulocytes are high, the platelet count is normal, the direct antiglobulin test is negative, and the urine contains hemoglobin and hemosiderin. Write the complication, the structural defect most likely behind it, and the mechanism in one sentence. Then take each laboratory abnormality in turn and write which step in the destruction of red cells it marks. Write why the platelet count and the direct antiglobulin test matter here, naming the three conditions that also produce schistocytes and the one that would also destroy red cells immunologically. Write why she may become iron deficient despite ongoing hemolysis, naming where the iron is lost. Then a second patient develops fever 6 weeks after valve replacement, with a new paravalvular leak and heart block. Write where this infection starts and where it spreads, the organism you expect, the structure on the prosthesis it depends on and what that structure does to host defences, and the four laboratory features that identify that organism. Finish by writing how late disease after valve replacement differs in its organisms.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-04-C10-Q04

Anemia three years after valve replacement

A 58-year-old woman comes to the physician because of 4 months of fatigue and intermittently dark urine. She underwent mechanical mitral valve replacement 3 years ago. Temperature is 36.7 °C (98.1 °F). Hemoglobin is 8.6 g/dL, the platelet count is 240,000/mm3, and the blood film shows fragmented red cells including helmet cells. Lactate dehydrogenase is high, unconjugated bilirubin is raised, haptoglobin is undetectable, and the direct antiglobulin test is negative. Urinalysis shows hemoglobin and hemosiderin with no red cells. Transesophageal echocardiography shows a gap between the sewing ring and the annulus with a high-velocity jet through it. Which of the following best explains her anemia?

A. Destruction of red cells by warm autoantibodies directed against the red cell membrane

B. Shear fragmentation of red cells by a high-velocity jet through a paravalvular leak

C. Consumption of platelets and fibrin strand formation in the microvasculature

D. Hepcidin-driven iron restriction from chronic inflammation on the prosthesis

Why did you pick it?